

ZAYAAN BHANWADIA

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EDUCATION

University of Toronto

Sept. 2025 – Dec. 2028

Hon. B.Sc. in Computer Science – Software Engineering Specialist (Co-op) | GPA: 3.7/4.0

Toronto, ON

Activities: 3x Hackathon Winner, Web Developer @ UTRA, VP Technology @ CS Student Association

EXPERIENCE

Incoming Full Stack Software Developer Intern

Sept. 2026 – Dec. 2026

FreshBooks

Toronto, ON

- Joining a product team to ship production code across **Python/Ruby** microservices and **React** UI for **10M+** users.

Autonomy Software Engineer (*Mapping & Planning*)

July 2026 – Present

University of Toronto Formula Racing – Driverless

Toronto, ON

- Developed mapping and path-planning software in C++ and ROS 2 for **Canada's first driverless FSAE car**.
- Built a **SLAM** pipeline that maps unknown tracks to within **18 cm** using particle filtering and graph optimization.
- Redesigned the path planner to navigate sharp turns that stalled the legacy system, with **96%** of plans on track in sim.
- Enabled **LiDAR**-only planning, removing the camera neural network to cut **~20 ms** of latency per planning cycle.

Software Engineer Intern (*Backend AI*)

July 2026 – Present

FlyRank AI

Toronto, ON

- Built a centralized embeddable widget platform, routing form submissions to a dashboard via a one-line script tag.
- Secured a public endpoint with validation, rate limiting, and geolocation checks, cutting spam by **80%**.
- Prevented duplicate charges under **50 concurrent requests** by deduplicating billable events at the database level.
- Synced Stripe payment updates through secure, idempotent webhooks, eliminating duplicate records and manual fixes.

Full Stack Software Developer Intern

May 2026 – July 2026

KorraNet Creative

Remote

- Optimized Meta OAuth token handling and storage, reducing token refresh failures by **40%+** for AI integrations.
- Reduced system crashes and failed API calls by **85%+** by adding API error handling and timeout prevention.
- Migrated hardcoded secrets to Firebase Secret Manager and implemented real-time secret ingestion for API calls.

PROJECTS

Embedded Key-Value Store | C++, POSIX Threads, Linux

- Built an embedded C++ database with data sharding to prevent global lock bottlenecks and improve throughput.
- Implemented shared locks for non-blocking reads across **4 threads**, preventing race conditions during writes.
- Created a file compaction routine to safely clean up keys, **reducing disk usage by 45%+** without pausing the app.
- Designed data persistence by writing updates to a log file before memory to ensure crash recovery and durability.

Context Sync Extension | TypeScript, VS Code Extension API

- Developed a VS Code extension with **600+ downloads** that automates AI chat context sharing across environments.
- Implemented Dijkstra's search algorithm to select sessions referenced, resulting in a **25%+** reduction in token usage.
- Designed a graph-based Markdown schema to maximize context density per token across AI assistant sessions.
- Integrated GitHub Copilot, OneDrive, and Google Drive to automate syncing and enable one-click AI workspace setup.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C++, C, SQL, Bash, HTML/CSS

Libraries/Frameworks: FastAPI, Express.js, Node.js, SQLAlchemy, React.js, ROS 2, OpenCV, YOLOv11

Developer Tools: Linux, Docker, Git, GitHub, Claude Code, pgAdmin, VS Code, Jira/Confluence

Cloud/Databases: PostgreSQL, MySQL, Google Cloud Platform, Microsoft Azure, IBM Cloud, Supabase